

Brandon Imstepf

Ph.D. Candidate
Applied Mathematics Program
University of California, Merced

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Research Interests

Mathematical and Computational Neuroscience, Bioinformatics, Machine Learning, Data Science

Education

Anticipated Spring 2027 **University of California, Merced**, *Doctor of Philosophy*, Applied Mathematics
Doctoral Advisor: Dr. Suzanne Sindi
Spring 2020 **University of California, Santa Cruz**, *Bachelor of Arts*, Mathematics
Concentration: Computational

Project & Research Experience

Fall 2018 - **Slugbotics**, *University of California, Santa Cruz*, MATE Underwater ROV Challenge
Spring 2019 Assisted in the design and construction of an underwater remotely controlled submersible vehicle.
Summer 2022-Present **Graduate Student Researcher**, *University of California, Merced*
Collaborative effort with Sindi Lab to research neurodegenerative diseases.
Summer 2024-Present **Sindi-Raj Lab Collaboration**, *University of California, Merced - University of California, San Francisco*
Collaborative effort between Sindi Lab and Ashish Raj's group of the Department of Radiology to study the transport of pathological Tau proteins in Alzheimer's Disease.
Summer 2025-Present **Joint Genome Institute (JGI) Internship and Extension**, *Lawrence Berkeley National Lab*
Summer internship followed by the New Energy Science Workforce (RENEW) Initiative, wherein I work with Brian Bushnell developing machine learning algorithms for applications on genomics big data, outperforming deterministic approaches for gene calling.

Relevant Classwork

Spring 2023 **Frontiers of Molecular and Cellular Biotechnology**
Fall 2023 **Numerical Solutions of Ordinary Differential Equations**
Spring 2024 **Numerical Solutions of Partial Differential Equations**
Fall 2024 **Bioinformatics**
Spring 2025 **Computation and Modeling for Interdisciplinary Biophysical Sciences, Biomaterials and Biotechnology**

Publications & Presentations

- Feb 2023 **A First Year's Perspective on Mathematical Modelling**
Invited Presentation, University of California, Merced
- Sep 2023 **Predicting Atrophy in Alzheimer's Via Network Laplacian**
Invited Presentation, Mathematical Biology Seminar, University of California, Merced
- July 2024 **Bridging the Scales: From Protein to Brain in Alzheimer's Disease**
Invited Presentation, Korean Society for Mathematical Biology - Society for Mathematical Biology 2024, Konkuk University, Seoul
- October 2024 **Bridging the Scales: From Protein to Brain in Alzheimer's Disease**
Invited Presentation, Society for Industrial and Applied Mathematics - Northern and Central California 2024
- July 2025 **Machine Learning Applications in Gene Annotation**
Research Presentation - Lawrence Berkeley National Lab
- July 2025 **Accelerating Solutions of Nonlinear PDEs Using Machine Learning: A Case Study with the Network Transport Model**
Contributed Talk - Society for Mathematical Biology 2025, Edmonton

Honors & Awards

- Summer 2022-
Summer 2024 **Graduate Research Fellowship**, *National Institute of Health G-RISE Interdisciplinary Biomedical Science and Technology Program*, University of California, Merced
- Summer 2022 **Competitive Edge Summer Bridge Program Participant**, *UC Merced Graduate Division*, University of California, Merced
- Fall 2023,
Spring 2024,
Summer 2024 **Distinguished Service in Support of Applied Mathematics**, *UC Merced Graduate Division*, University of California, Merced
- Fall 2024,
Fall 2025 **Research and Training Grant Recipient**, *UC Merced Graduate Division*, University of California, Merced

Teaching & Work Experience

- 2018 **Mathematics Tutor**, *Santa Monica College*
Provided tutoring for math subjects ranging from pre-algebra to advanced calculus.
- Summer 2024-Present **Teaching Assistant**, *University of California, Merced*
Led discussion sections, formulated and administered quizzes for various undergraduate courses.
- May 2025-Present **Research Affiliate**, *Lawrence Berkeley National Lab*
An internship under Brian Bushnell designing and implementing neural networks in Java that outperform deterministic algorithms in gene calling.

Professional & Leadership Experience

- Fall 2023-Present **Applied Math Graduate Student Leadership Committee**, *UC Merced Graduate Division*, University of California, Merced
I meet monthly with a group of other Applied Mathematics graduate students and graduate program chair and vice chair. We discuss student concerns, departmental activities and other matters of importance to the graduate program.

Fall 2023-Fall 2024 **School of Natural Sciences Graduate Student Council**, *UC Merced Graduate Division*, University of California, Merced

I met once per semester with the Dean of Natural Sciences and representatives from other SNS programs. We provided input on campus-wide activities and the future direction of the School of Natural Sciences.

Software Skills

L^AT_EX, Python/Jupyter, MATLAB, R/RMarkdown/Quarto, Java, C++, Tableau, SQL, Regex, LLMs

Memberships

Society for Mathematical Biology

Society for Industrial and Applied Mathematics

References

Suzanne Sindi, PhD, *University of California, Merced*, Doctoral Advisor
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Ashish Raj, PhD, *University of California, San Francisco*, Primary Collaborator
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